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AI Banking Agent for processing consumer complaints

This document describes details, features and architecture for potential implementation of AI banking agents for automatic processing consumer complaints.

Use cases

Automated Data Extraction

The AI agent automatically identifies and extracts critical information from a submitted complaint (such as account numbers or personal details). AI agents needs to log every extraction with a link to full message content.

Smart Complaint Categorization

An AI agent analyzes the collected information to assign the complaint to the correct category. Categorization is done with directions coming from extensive *complaint tree* that is regularly updated to match bank's best practices and current state of law.

AI agent needs to keep always up-to-date version of complaint tree within context. Automatically assigned category is not final by definition and can be overridden by human. For the early stage of agent implementation each categorized item can be saved in a queue waiting for human approval.

Information Retrieval for Decision Support

A specialized AI agent connects to various internal banking systems to pull all necessary data and documentation required to resolve a specific complaint, so as to ensure preparation of a data-driven recommendation for the final decision.

Every operation performed on customer's data via banking APIs needs to be properly documented, with the reason why each request is made

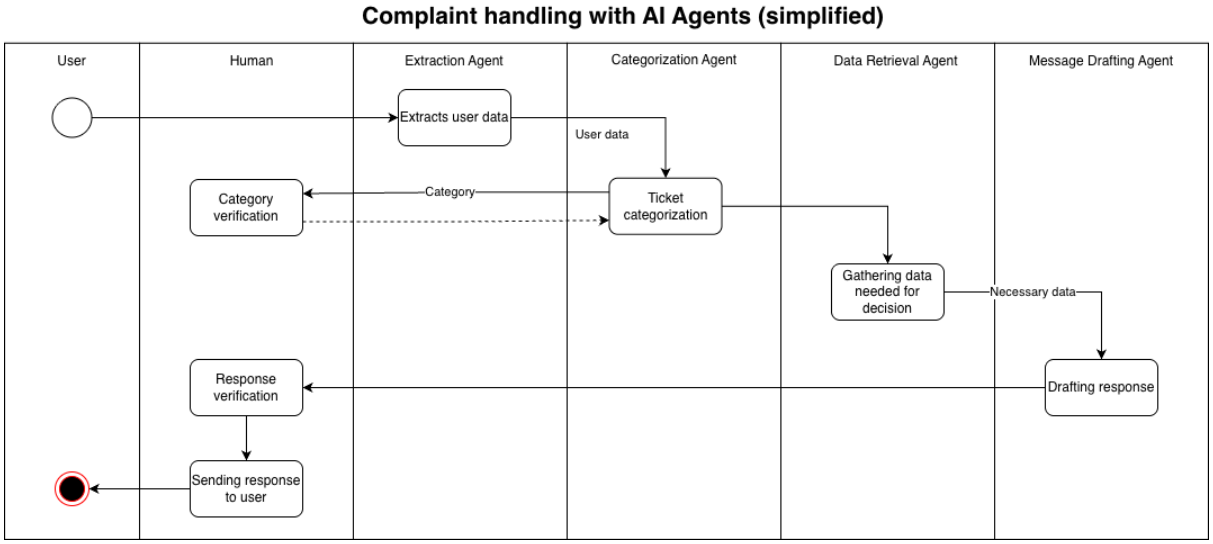
Drafting of Customer Response with decision

The AI agent generates a formal draft of the response letter to be sent to the customer. The draft includes detailed explanations and specific references to the documents and evidence used as the basis for the resolution.

Drafted message must be verified manually by human before sending to actual customer.

Technical perspective

Agent flow diagram



The diagram shows only flow simulation between 4 main agent, human (bank employee) and user. Entire can be linear like displayed on the diagram or can be orchestrated with another AI agent. To simplify this document, I'll focus on linear approach with no Agentic AI orchestration layer. Decision whether to finally implement orchestrating AI agent should be based on how complex process is and whether another layer of AI-based automatically orchestration is needed.

The diagram does not show integration with other system services.

Suggested tech stack

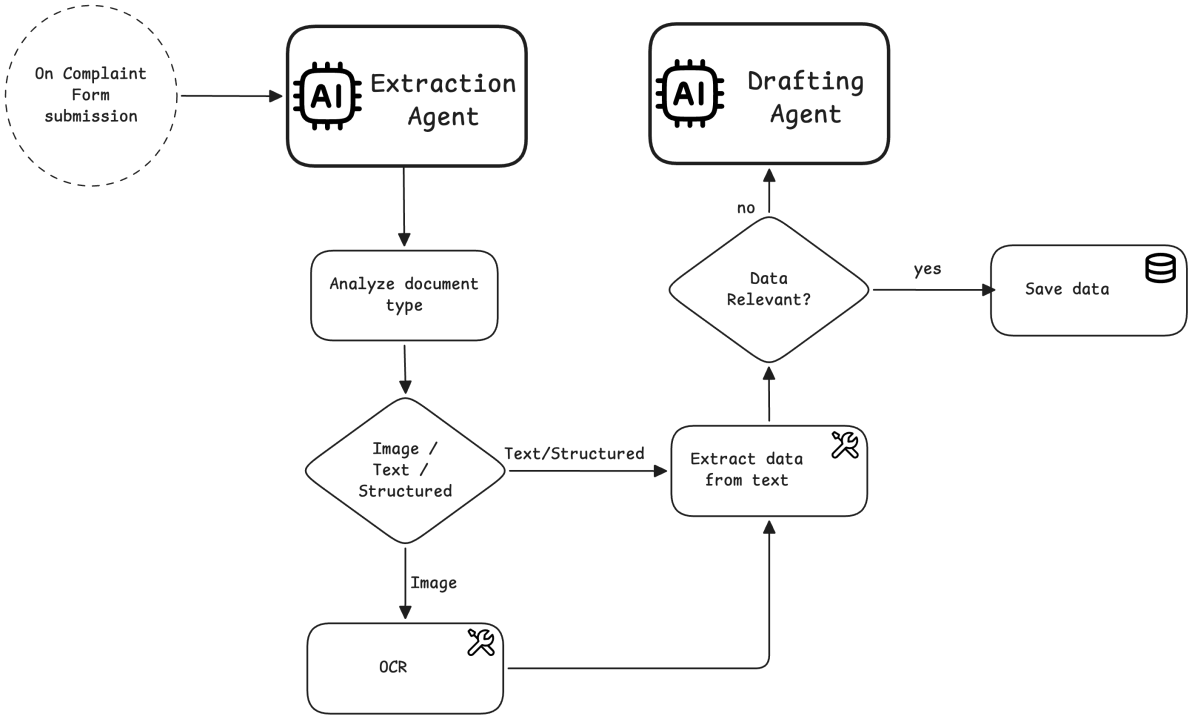
- LangGraph - low level agent orchestration framework
- LangChain - tools, retrievers, structured output, document loaders
- Langfuse - tracing, eval, prompt management, cost monitoring
- LLMs
    - Claude 4.6 Opus – For the most complex tasks
    - GPT 5.4 Mini – cheaper alternative
    - GLM 5.1 / MiniMax M2.7 / Kimi 2.5 / Qwen 3.5 – when considering self-hosting models
- LLM providers:
    - Enterprise API (Amazon Bedrock, Azure AI)
    - Self-hosted models via vLLM
- PostgreSQL
- Redis
- N8N – PoC

LLM providers

When choosing LLM providers into consideration should be taken costs, simplicity, maintenance efforts and law regulations regarding storing, parsing and using bank client's data.
The starting point solution for this decision can be enterprise grade APIs from AWS, Azure or GCP which ensures data privacy and should be complaint ready. As project grows it's worth to consider switching to self-hosted models what fully resolves privacy issues and minimizes costs, however, adds up maintenance efforts.

Agents' definitions

Extraction Agent



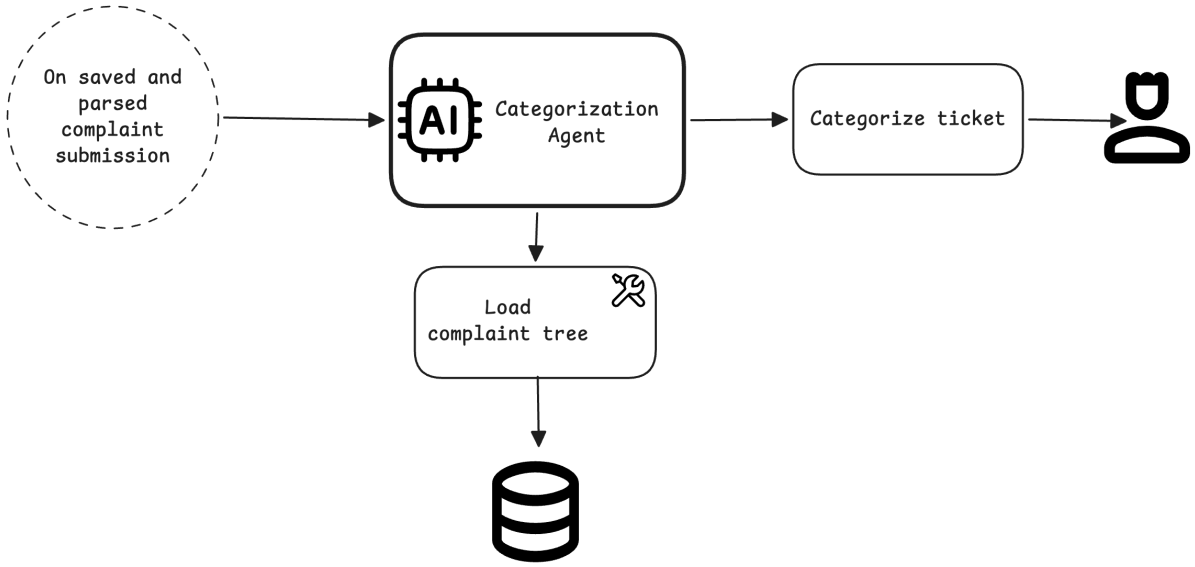
Extraction Agent is the first agent in the agentic chain of the process as it's working directly on data sent by the customer via complaint form. I assume that the form apart from text content can consist of appendix, which most probably will be docs, PDF files or images.

Firstly, agent should classify document based on its type. If document is structured (e.g. XML, JSON, docx or properly generated PDF) its content can be analyzed by each LLM. In case of image document (e.g. png, jpeg or PDF with scanned pages), the content of document needs to be converted into tokenizable text content via OCR tool.

OCR tool usage preceding extracting data from text can be replaced with multimodal LLM usage with decent vision capabilities to read the data.

After reading data from the form, AI agent should verify if data are relevant and complete. If yes, data can be saved and process can move on. Otherwise, the flow goes to drafting agent to prepare message for incomplete data, or any other decision maker for that process if relevant.

Categorization Agent



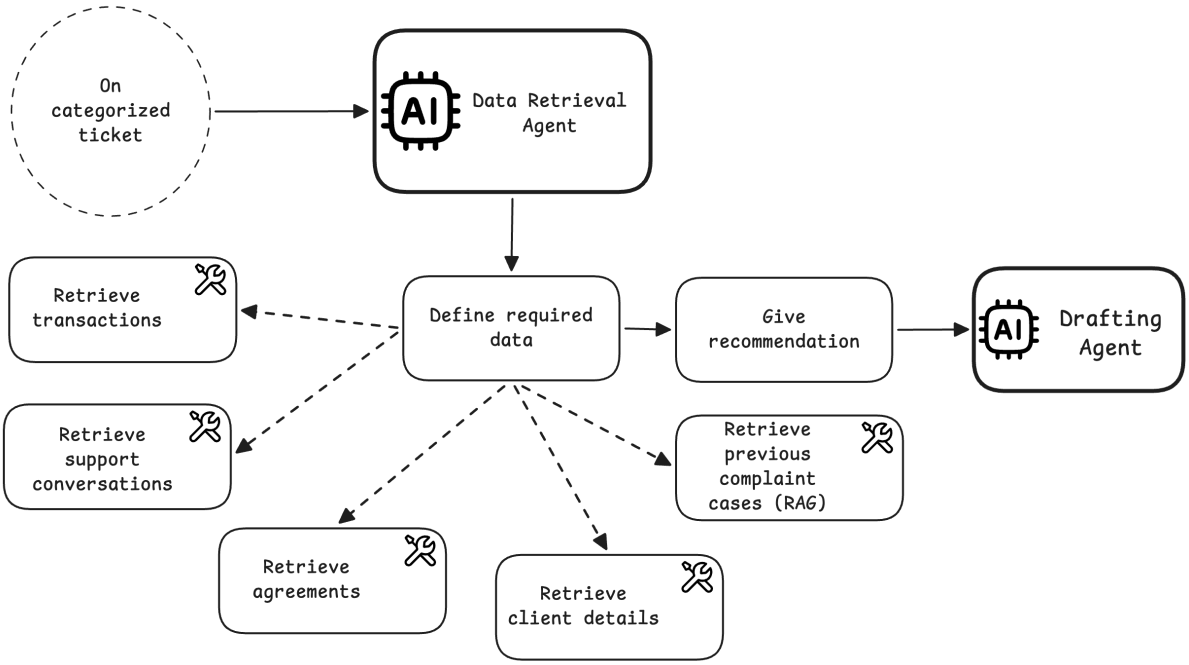
Categorization agent is second agent in the process that takes care of assigning proper category to each complaint related ticket. Agent is taking advantage of externally maintained complain tree database.

Entire complain tree content is loaded into AI chat context so that agent has direct and quick access to it. Because of that, complaint tree must be used in an LLM optimized format, which can be done in external database, or inside `load complaint tree` tool.

After ticket categorization data are forwarded to bank employee for verification or saved to a database for potential future verification (or no action).

Categorization agent is the simplest one, which with slightly modified architecture can be just a tool, instead of separate agent.

Data Retrieval Agent



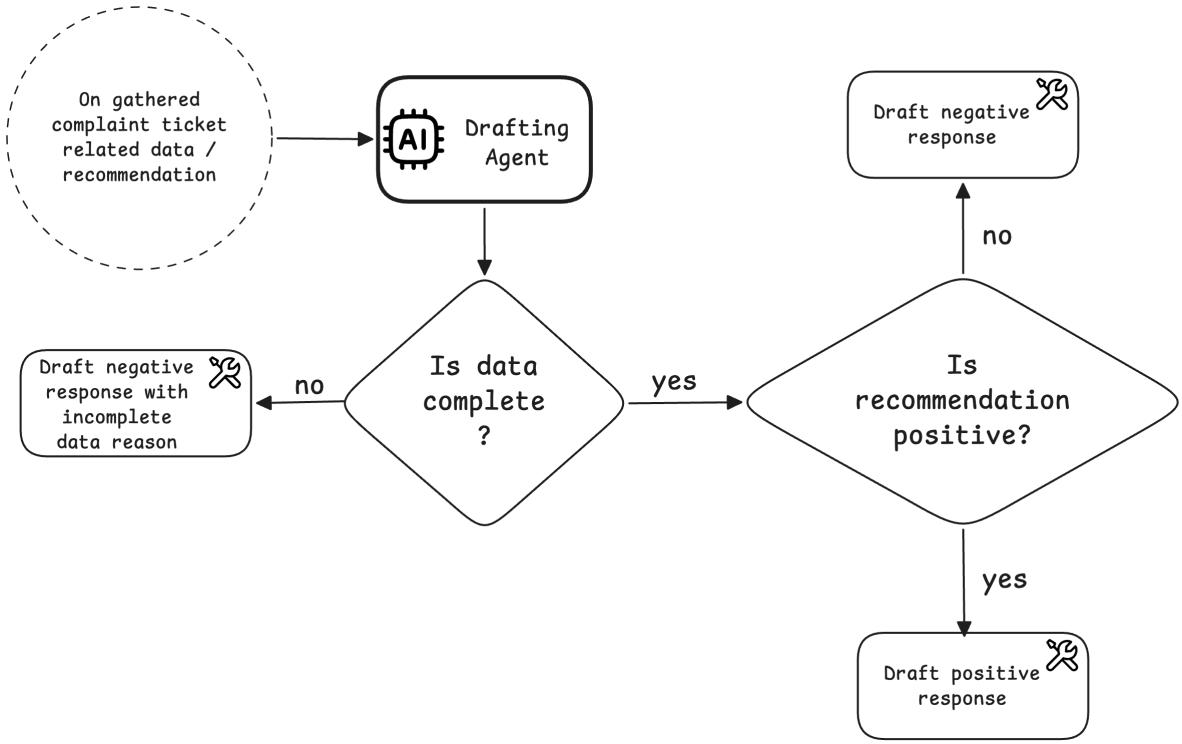
Data Retrieval Agent flow is the least linear. All of agent's action depends on specific complaint ticket category, information provided by client, type of client etc.

When agent receives categorized ticket (assuming here that the ticket categorization is accepted at this point) agent needs to define what data is required to decide for this case. This agent's action is kind of a blackbox, while we don't know based on what criteria agent can make this decision. This action probably requires loading some rules into context from source that keeps decision flow up to date. Every data needed can be retrieved with one of available tools connected to other system databases.

Another blackbox action is "give recommendation", where AI agent needs to decide whether to accept the complaint or not. Such action should be done with decision making rules defined inside prompt or any other external database to be injected in the context to ensure its relevance.

Recommendation generated in a previous step is then passed to Drafting Agent, most likely with needed customer data attached.

Message Drafting Agent



Message drafting agent is the last agent in the entire process that has a simple job of drafting a response that in case of positive human verification can be send to the client.

Drafting agent job can be launched at the very end of process where there is already a recommendation, or earlier when other agent (most likely extraction agent) reported missing data. Agent job can be also started on another potential failures during the process, if such failure requires contacting client.

Based on provided context:
- Data complete or not
- Process successful so far or not
- Recommendation: positive or not

Is drafting a response using attached customer data. The response should be saved to production system used to client communication in draft mode, where employee can verify, change if needed and finally send response.

In case of simple drafting process, this agent can also be swapped with a simple AI-based tool that based on input (recommendation, reason, customers data) can generate a draft.

Reporting and logging activity

For financial institutions, full auditability of AI system actions is not optional - it is a regulatory requirement. The complaint handling system is designed to ensure every decision made by any agent is traceable, explainable, and compliant with the EU AI Act as well as applicable Polish financial regulations (including KNF guidelines and DORA requirements).

What will be logged

Every action performed by each of the four agents - Extraction Agent, Categorization Agent, Data Retrieval Agent, and Message Drafting Agent – will be recorded in a structured log entry containing:

- Timestamp - exact date and time of the action
- Agent identity - which agent performed the action
- Action type - what the agent did (e.g. data extraction, ticket categorization, response drafting)
- Input context - what data the agent received
- Reasoning process - the chain of thought used to reach the outcome
- Output - the result produced by the agent
- Human verification events - category and response verification steps performed by a human operator

This level of detail ensures that any decision can be fully analyzed and justified after the fact.

Tooling

Two complementary layers can handle observability:

Langfuse can used as the primary tracing and monitoring platform. It provides a real-time dashboard for tracking agent runs, token usage, and errors across the entire pipeline. Each agent invocation creates a trace in Langfuse, making it easy to inspect the full execution path of any complaint handling session.

Structured database logging can serves as the persistent audit record. All log entries are written to a database in a defined, consistent schema - ensuring long-term retention, queryability, and the ability to generate compliance reports on demand.

Together, these two layers provide both operational visibility for the engineering team and a defensible audit trail for regulators and compliance officers.

Alerting and Anomaly Detection

The system should not rely only on passive logging, but it should actively monitor agents behavior and raises alerts in case of an anomaly detected.

Automated alerts should be triggered when:
- A single agent category accounts for an unusual share of processed complaints within a given time window (potential miscategorization drift)
- Response drafting latency exceeds defined thresholds, indicating a possible bottleneck or model degradation
- Human verification consistently overrides a specific agent's output, suggesting the agent may need retraining or prompt adjustment
- Any agent produces an error rate above a set baseline

Alerts should be routed to the responsible engineering or operations team in real time, enabling fast diagnosis before issues affect customers or accumulate into compliance risks.

Business Reporting and Time Savings

Beyond regulatory compliance, the data captured through logging powers practical business intelligence. Because every complaint session and its each step is recorded, the system can generate reports that directly measure operational efficiency and return on investment. Potential metrics:

- Time saved per complaint - The most direct metric. By comparing the average handling time before and after system deployment, the bank can quantify exactly how many minutes per complaint are saved through AI-assisted processing.
- Volume capacity - Logs reveal how many complaints are processed per day and per agent. This data shows at what scale the system operates without requiring additional human resources
- Category breakdown and complaint trends - Aggregated categorization data shows which complaint types are most frequent, how their volume changes over time, and whether specific products or processes are generating disproportionate issues. This gives management early visibility into operational problems before they escalate.
- Human interaction / override rate - Tracking how often human operators modify or reject agent outputs provides a quality signal over time. A declining override rate indicates the system is improving. A sudden spike indicates something has changed and needs attention.
- Response time per complaint type - Handling time broken down by category. This helps identify which complaint types take longest to resolve and where process improvements would have the highest impact.

Reference documents

1. https://www.velobank.pl/klienci-indywidualni/biuro-prasowe/rewolucja-w-obszarze-reklamacji-velobank-znowu-wyznacza-trendy.html
2. SpeedNet Report, “Beyond AI: W stronę Agentic Banking”